MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 6**

Inverse functions and exponential functions, and their derivatives (Sections 6.1–6.2, Lectures 11–12); logarithms, their derivatives and logarithmic differentiation (Sections 6.3–6.4, Lecture 13).

Key facts to have at hand (Lectures 11–13)

- **One-to-one:** f never takes the same value twice ($x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$; horizontal line test). Then f has an *inverse*: $f^{-1}(y) = x \iff f(x) = y$. The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f. *Undoing:* $f^{-1}(f(x)) = x$ for every x in the domain of f, and $f(f^{-1}(y)) = y$ for every y in the range of f. To find f^{-1}: write $y = f(x)$, solve for x, and keep track of the range. The graph of f^{-1} is the graph of f reflected in $y = x$. And f^{-1} is *not* $1/f$.
- **The derivative of an inverse:** if f is one-to-one and differentiable on an open interval and $f'\big(f^{-1}(a)\big) \neq 0$, then $(f^{-1})'(a) = \dfrac{1}{f'\big(f^{-1}(a)\big)}$ — f' is taken at $f^{-1}(a)$, not at a. If $f'(b) = 0$, then f^{-1} has a vertical tangent at $a = f(b)$ and no derivative there.
- **Exponential functions** ($b > 0$): $b^{x+y} = b^x b^y$, $b^{x-y} = b^x / b^y$, $(b^x)^y = b^{xy}$, $(ab)^x = a^x b^x$. For $b > 1$: increasing, $b^x \to \infty$ as $x \to \infty$, $b^x \to 0$ as $x \to -\infty$; for $0 < b < 1$ the other way round; always $b^x > 0$. e is the base whose slope at 0 is 1: $(e^x)' = e^x$; with the chain rule $\big(e^{g(x)}\big)' = e^{g(x)} g'(x)$. e^x is increasing and continuous, with range $(0, \infty)$; $\lim_{x \to -\infty} e^x = 0$, $\lim_{x \to \infty} e^x = \infty$.
- **Logarithms** ($b > 0$, $b \neq 1$): $\log_b x = y \iff b^y = x$ ($x > 0$). Undoing: $\log_b(b^x) = x$ for every x, $b^{\log_b x} = x$ for $x > 0$. Laws ($x, y > 0$): $\log_b(xy) = \log_b x + \log_b y$, $\log_b \frac{x}{y} = \log_b x - \log_b y$, $\log_b(x^r) = r \log_b x$ — and no law for $\log_b(x + y)$. $\ln = \log_e$; $\log_b x = \dfrac{\ln x}{\ln b}$; $b^x = e^{x \ln b}$. For $b > 1$: $\lim_{x \to 0^+} \log_b x = -\infty$, $\lim_{x \to \infty} \log_b x = \infty$.
- **Derivatives:** $(\ln x)' = \dfrac{1}{x}$ ($x > 0$); $(\ln |x|)' = \dfrac{1}{x}$ ($x \neq 0$); $\big(\ln |g(x)|\big)' = \dfrac{g'(x)}{g(x)}$ where $g(x) \neq 0$; $(b^x)' = b^x \ln b$; $(\log_b x)' = \dfrac{1}{x \ln b}$. *Laws first*: expand a logarithm of a product, quotient or power before differentiating.
- **Is the base constant, or the exponent?** $(b^n)' = 0$; $\big([f(x)]^n\big)' = n[f(x)]^{n-1} f'(x)$; $\big(b^{g(x)}\big)' = b^{g(x)} \ln b \cdot g'(x)$; for $[f(x)]^{g(x)}$ ($f(x) > 0$): *logarithmic differentiation* — take ln of both sides, expand by the laws, differentiate implicitly, solve for y' — or write it as $e^{g(x) \ln f(x)}$. (Logarithmic differentiation also turns products and quotients into sums; where y can be negative, take $\ln |y|$.) And $e = \lim\limits_{x \to 0} (1 + x)^{1/x} = \lim\limits_{n \to \infty} \left(1 + \dfrac{1}{n}\right)^n$.

Part I · Inverse and exponential functions (6.1–6.2) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is one-to-one, then $f^{-1}(x) = \dfrac{1}{f(x)}$.
2. The function $f(x) = x^2 + 1$, $x \geq 0$, is one-to-one.
3. $\dfrac{d}{dx} e^{3x} = e^{3x}$.
4. The function $f(x) = x^4$ has an inverse function on $\mathbb{R}$.
5. If f is one-to-one, the domain of f^{-1} is the range of f.
6. If f is one-to-one and differentiable on $\mathbb{R}$, $f(3) = 7$ and $f'(3) = 4$, then $(f^{-1})'(7) = \dfrac{1}{4}$.
7. $(f^{-1})'(a) = \dfrac{1}{f'(a)}$ for every one-to-one differentiable function f.
8. $e^{x+y} = e^x + e^y$ for all real x and y.
9. $\lim\limits_{x \to -\infty} 3^x = 0$.
10. $\dfrac{d}{dx} e^2 = e^2$.

Problem 1 *Inverse functions: one-to-one, the formula, the domain and the range.*

For each function: show that it is one-to-one, find a formula for its inverse, and give the domain and the range of the function and of its inverse.

(a) $f(x) = \dfrac{2x-1}{x+3}$

(b) $g(x) = x^2 + 4x, \;\; x \geq -2$ (complete the square)

(c) $h(x) = \sqrt{5-2x}$

Problem 2 *The derivative of an inverse function — without a formula for the inverse.*

(a) Let $f(x) = x^5 + 2x + 3$. Show, without derivatives, that f is increasing (hence one-to-one). Find $f^{-1}(6)$ and $(f^{-1})'(6)$, and also $(f^{-1})'(3)$.

(b) A function f, one-to-one and differentiable on $\mathbb{R}$, has the values below. Find (i) $(f^{-1})'(5)$; (ii) $(f^{-1})'(2)$; (iii) the equation of the tangent line to $y = f^{-1}(x)$ at $x = 9$.

x	0	2	5
$f(x)$	2	5	9
$f'(x)$	4	3	$\frac{1}{2}$

(c) Let $f(x) = (x-1)^3 + 2$. At which number a is f^{-1} *not* differentiable, and why? Check with a formula for f^{-1}.

Problem 3 *Limits of exponential functions.*

(a) $\displaystyle\lim_{x\to\infty} \frac{3e^{2x}+5}{e^{2x}-4}$ and $\displaystyle\lim_{x\to-\infty} \frac{3e^{2x}+5}{e^{2x}-4}$

(b) $\displaystyle\lim_{x\to 3^+} 2^{1/(x-3)}$ and $\displaystyle\lim_{x\to 3^-} 2^{1/(x-3)}$

(c) $\displaystyle\lim_{x\to\infty} 5^{-x}\sin x$

(d) $\displaystyle\lim_{x\to\infty} \frac{2^x+3^x}{3^x-2^x}$

Problem 4 *Derivatives with e^u.*

Differentiate. For every $e^{\cdots}$ say what the *inside* is before you write anything.

(a) $y = e^{x^3-3x}$

(b) $y = 2x\,e^{-x/2} + e^{-x/2}$

(c) $y = \dfrac{e^{2x}}{x+3}$

(d) $y = \left(e^x + 3\right)^4$

(e) $y = e^{2x}\cos 3x$

(f) $y = e^{e^x}$

Problem 5 *Horizontal tangents and tangent lines.*

(a) Find the points of the curve $y = (x^2-3)e^x$ at which the tangent line is horizontal.

(b) Find the equation of the tangent line to $y = (x+1)e^{-2x}$ at $x = 0$.

(c) For which constants k does $y = e^{kx}$ satisfy $y'' - 5y' + 6y = 0$ for every x?

Problem 6 *Graphs of exponential functions (homework).*

Without a calculator, sketch each graph and state the domain, the range and the horizontal asymptote(s); start from $y = e^x$ or from $y = b^x$ for a suitable b, and say which transformations you used.

(a) $y = 5^{-x}$

(b) $y = 2 - e^{x+1}$

(c) $y = e^{-|x|}$

(d) $y = e^{2-x} - 1$

Part II · Logarithms; logarithmic differentiation (6.3–6.4) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. $\log_2 32 = 5$.
2. $\ln(a+b) = \ln a + \ln b$ for all $a, b > 0$.
3. $\log_3 \dfrac{1}{9} = -2$.
4. $\ln\left(e^x\right) = x$ for every real x.
5. $e^{\ln x} = x$ for every real x.
6. $\dfrac{d}{dx} \ln x = \dfrac{1}{x}$ for every $x \neq 0$.
7. For every $x \neq 0$, $\dfrac{d}{dx} \ln\left(x^2\right) = \dfrac{2}{x}$.
8. For every x, $\dfrac{d}{dx} 5^x = x\, 5^{x-1}$.
9. For $x > 0$, $\dfrac{d}{dx} x^\pi = \pi\, x^{\pi-1}$.
10. The function $\log_{1/2} x$ is increasing.

Problem 7 *Derivatives with* ln *— laws first.*

Differentiate. Where the laws of logarithms simplify the function, use them *before* you differentiate.

(a) $y = \ln\left(4 + x^4\right)$

(b) $y = \ln\left[(2x+1)^3 (x^2+4)^5\right]$

(c) $y = x^3 \ln(2x)$, $x > 0$

(d) $y = \ln|\sin 2x|$

(e) $y = \log_3\left(5x^2 + 1\right)$

(f) $y = \dfrac{\ln x}{x^2}$

Problem 8 *Logarithmic differentiation.*

(a) $y = \dfrac{\sqrt{x}\,(x+2)^5}{(x^2+4)^2}$, $x > 0$: find y', then $y(2)$ and $y'(2)$.

(b) $y = \dfrac{(x-3)^5 e^{2x}}{x^2+1}$: here y can be negative — take $\ln|y|$. Find y' and $y'(0)$.

(c) $y = \dfrac{(x+1)^4\sqrt{2x-1}}{x^2+3}$, $x > \frac{1}{2}$: find y', then $y(1)$ and $y'(1)$.

(d) $y = x^3(x-4)^2$: take $\ln|y|$ and find y'; then find *all* the points of the curve at which the tangent line is horizontal.

Problem 9 *Is the base constant, or the exponent?*

(a) $y = x^{\sqrt{2}} + \left(\sqrt{2}\right)^x$, $x > 0$

(b) $y = (2x)^x$, $x > 0$.

(c) $y = \left(1 + \dfrac{1}{x}\right)^x$, $x > 0$

(d) The equation of the tangent line to $y = (x+1)^x$ at $x = 2$.

Problem 10 *Equations with logarithms and exponentials.*

Solve. Check every candidate against the domain of the equation.

(a) $e^{3x-1} = 7$

(b) $\ln(x+2) + \ln(x-1) = \ln 10$

(c) $2^{x+1} = 5^x$

(d) $\log_2 x + \log_4 x = 3$

(e) $e^{2x} - 5e^x + 6 = 0$

Problem 11 *The number e as a limit; limits with logarithms.*

(a) Show that $\lim_{x\to 0}(1+3x)^{1/x} = e^3$. (Hint: the derivative of $\ln(1+3x)$ at 0.)

(b) Find $\lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n$.

(c) $\lim_{x\to\infty}\big[\ln(2x+1) - \ln(x+3)\big]$

(d) $\lim_{x\to\infty}\frac{\ln(x^4)}{\ln(x^2)}$

Problem 12 *Inverse functions meet logarithms (reserve).*

(a) Let $f(x) = x^3 + \ln x$, $x > 0$. Show that f is one-to-one, and find $(f^{-1})'(1)$.

(b) Let $g(x) = 2 + e^{2x-1}$. Find g^{-1}, with its domain and range, and find $(g^{-1})'(3)$ in two ways: from your formula, and from $\dfrac{1}{g'\big(g^{-1}(3)\big)}$.

(c) Use the formula for the derivative of an inverse function to show that $(\log_b x)' = \dfrac{1}{x\ln b}$, starting from $(b^x)' = b^x \ln b$ ($b > 0$, $b \neq 1$).

Challenge *(for those who finish early, or at home)*

1. Let $f(x) = \log_x 2$ for $x > 0$, $x \neq 1$ (the base varies!). Find $f'(x)$.
2. Differentiate $y = x^{x^x}$, $x > 0$ (the exponent is x^x).
3. Show that the equation $\ln x = 3 - x$ has exactly one solution, and find an interval of length $\frac{1}{2}$ that contains it.